Elderly Care Management System

Low-Fidelity Paper Wireframe & Prototype Design Specification

STUDENT DETAILS

Name: Farzana Sadia Takee
Student ID: 2211175
Course Project: Elderly Care Management System
Submission Type: Paper Wireframe / Low-Fidelity Design

DESIGN PHILOSOPHY

- **Senior-Centric Usability:** High contrast, large click targets, clear visual hierarchy.
- **Dual-Role Architecture:** Seamless handoff between Family Members and Caregivers.
- **Safety First:** Instant one-touch SOS and prominent medication alerts.

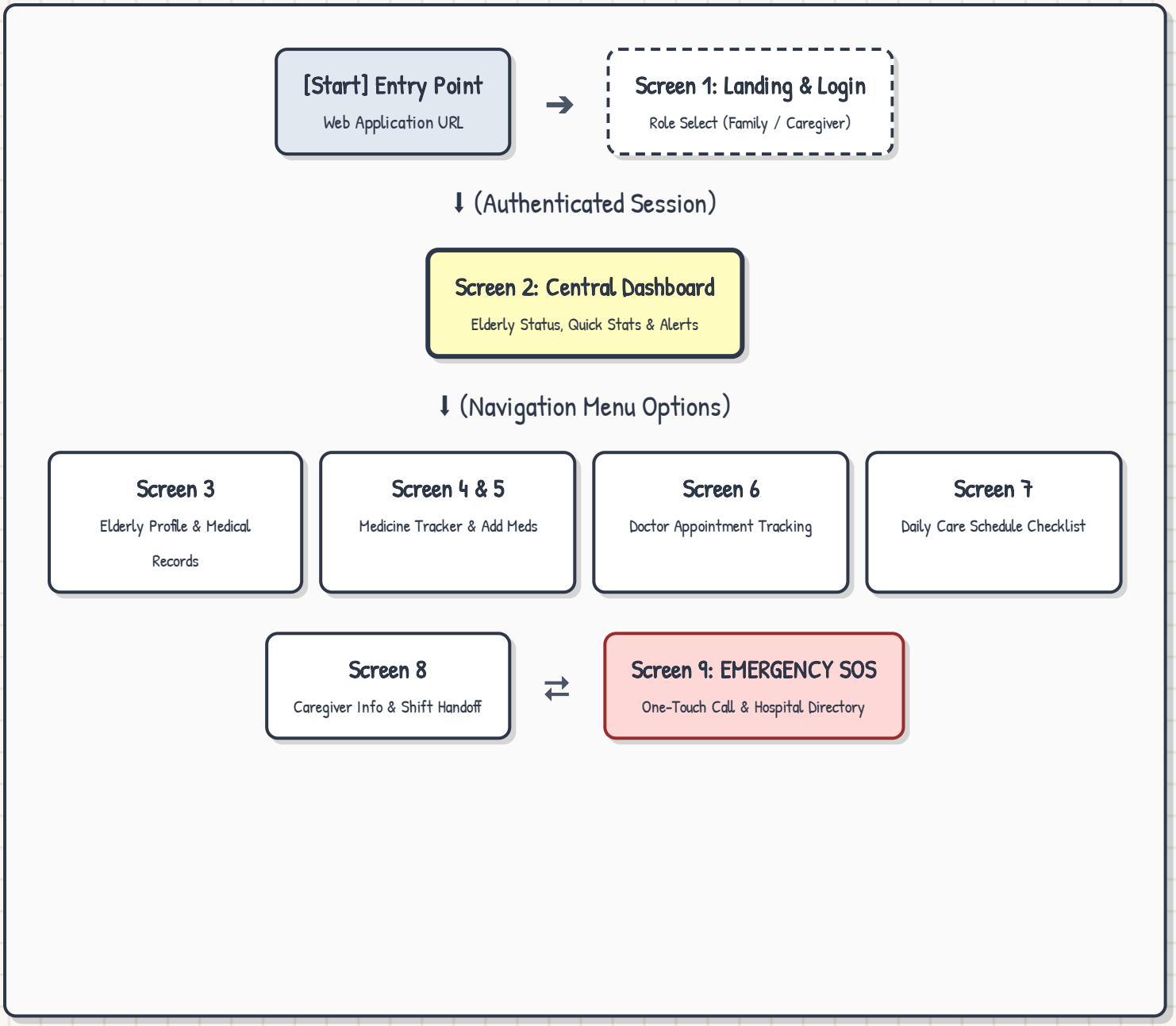
SEQUENCED INTERFACE INDEX (1 SCREEN PER PAGE)

1. Screen Flow & Sitemap Architecture
2. Landing & Role Authentication
3. Family Member Dashboard
4. Elderly Profile & Medical Details
5. Medicine Schedule & Tracker
6. Add / Edit Medication Form
7. Doctor Appointment Tracking
8. Daily Care Routine Checklist
9. Caregiver Info & Duty Notes
10. Emergency SOS Directory

01. Navigation Flow & System Architecture

Hierarchical map showing user navigation and screen transition sequence

Sequence Map



User Flow Rationale

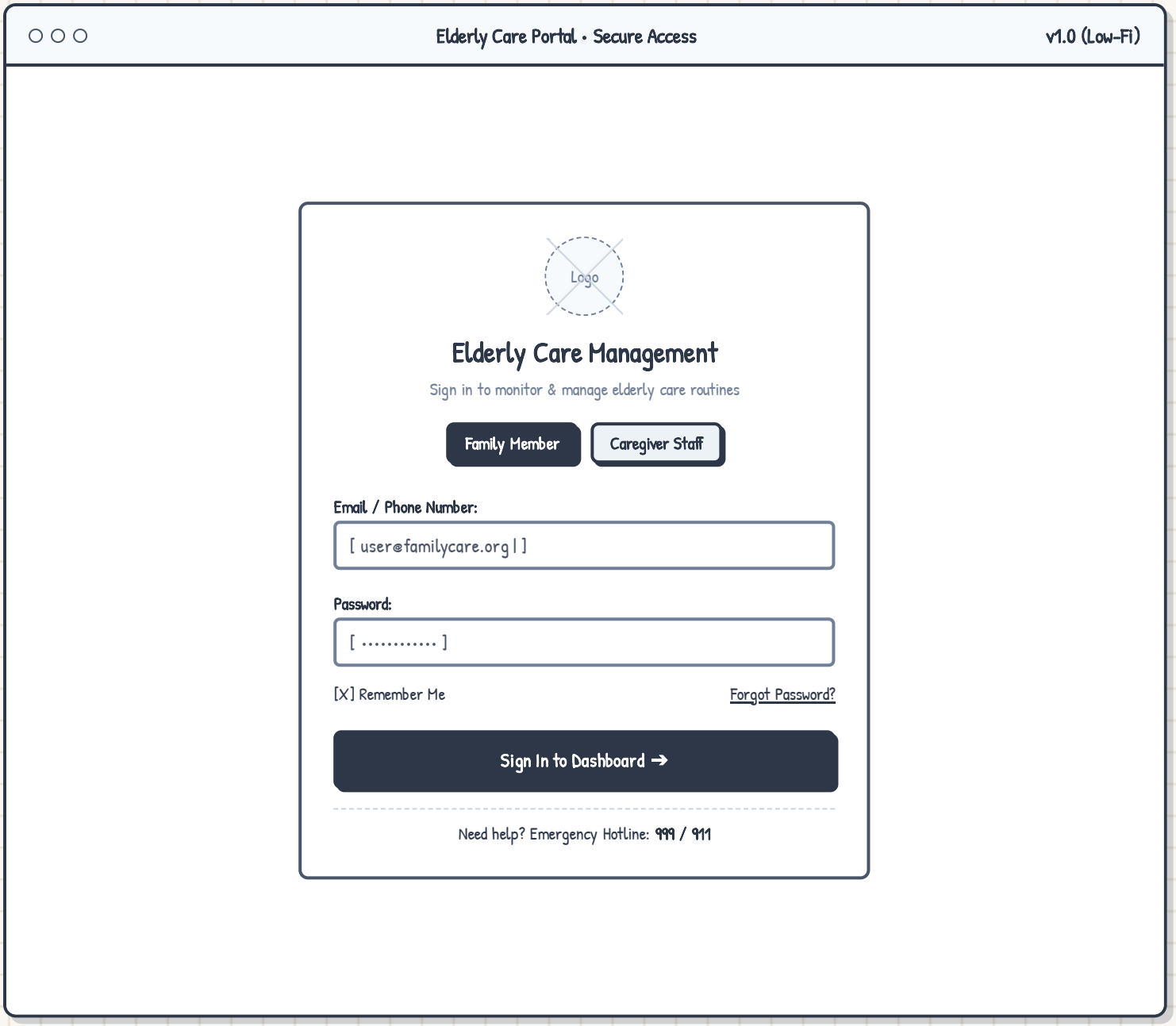
- **Zero Friction Login:** Family members and caregivers can immediately switch roles.
- **Hub & Spoke Model:** The Dashboard acts as the central command hub.
- **Persistent SOS:** Emergency trigger accessible from every single screen.

Target Personas

- **Primary:** Family Member (Son/Daughter) managing parent's care remotely.
- **Secondary:** Daily In-Home Caregiver executing medication & meal schedules.

02. Screen 1: Landing & Role-Based Login

Clean authentication portal with dual-role selector for Family and Caregiver



UI Wireframe Notes

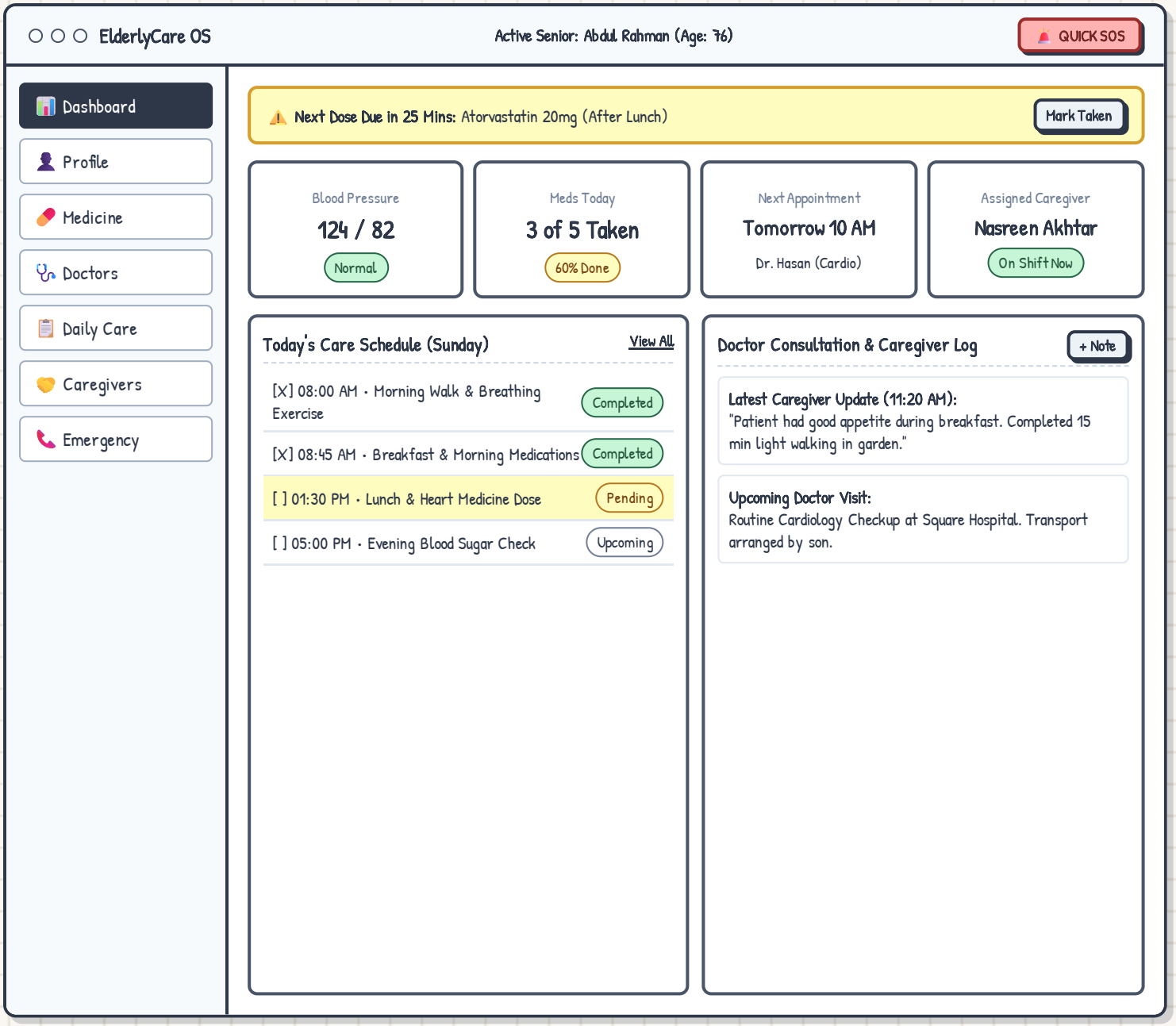
- **Role Toggle:** Instantly switches permissions (View/Edit rights vs Care Log updates).
- **High Accessibility:** Large input boxes for easy typing on touchpads or tablets.
- **Quick Recovery:** Password reset and emergency support numbers clearly visible.

User Actions

- Select user role.
- Enter credentials.
- Auto-route to Family Member Dashboard on submit.

03. Screen 2: Family Member Central Dashboard

High-level overview: active patient vitals, upcoming meds, care schedule, and rapid SOS



Dashboard UX Highlights

- **One-Glance Digest:** Family members immediately see if any medication is missed.
- **Vitals Cards:** Shows latest BP, blood sugar, and caregiver shift status.
- **Red SOS Button:** Positioned in header on every page for extreme emergencies.

Key Interactions

- Click 'Mark Taken' to record medication adherence.
- Click any nav item to jump to dedicated module.

04. Screen 3: Elderly Profile & Medical History

Comprehensive profile: personal data, chronic illnesses, blood group, allergies, and doctors

ElderlyCare OS > Elderly Profile

Edit Profile

- Dashboard
- Profile
- Medicine
- Doctors
- Daily Care
- Caregivers
- Emergency

Photo

Abdul Rahman Chowdhury

Age: 76 Years · Male · DOB: 14 Aug 1950 · Blood Group: B Positive (B+)

Home Address: House 42, Road 11, Dhanmondi, Dhaka

Allergy: Penicillin

Mobility: Needs Cane

Chronic Conditions

Type-2 Diabetes	Since 2012
Hypertension (High BP)	Since 2015
Mild Osteoarthritis	Knee Joint
Cataract (Post-Op)	Both Eyes

Primary Healthcare Contacts

Dr. K. M. Hasan
Cardiologist, Square Hosp
Call

Dr. Nusrat Jahan
Endocrinologist (Diabetes)
Call

Square Hospital Helpline
Emergency Desk
SOS

Special Care Directives

- Low-sodium diet mandatory.
- Hydration check every 2 hours.
- Keep walking stick beside bed at all times.
- Emergency contact son Tariq (01711-XXXXXX) if BP > 150/95.

Design Considerations

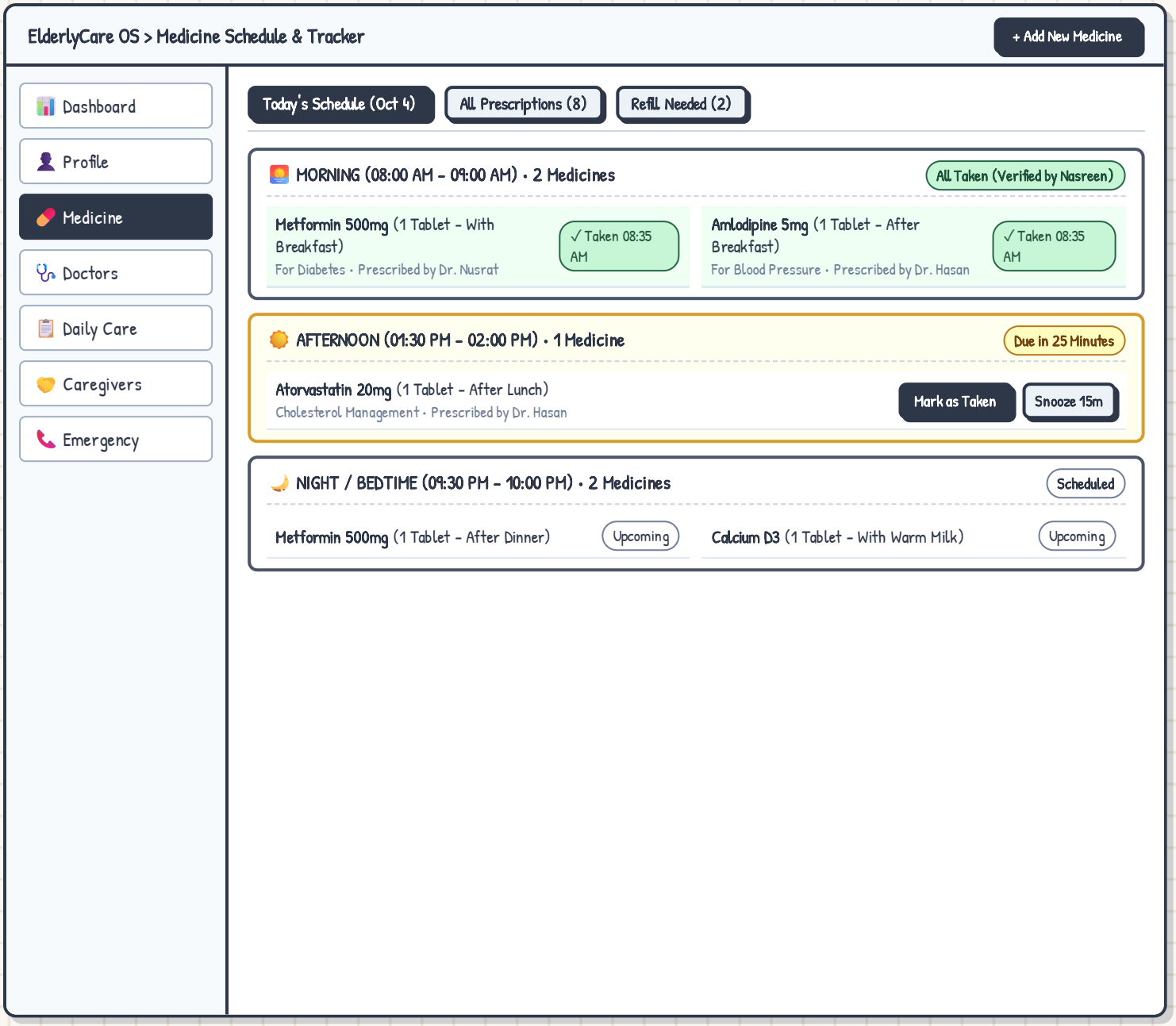
- **Critical Health Flags:** Blood group and Penicillin allergy stand out prominently.
- **Care Directives:** Clear instructions for newly arriving or rotating caregivers.
- **Quick Call Actions:** Fast dial buttons for primary treating physicians.

Data Handled

- Demographic & address details
- Medical conditions history
- Physician association records

05. Screen 4: Medicine Schedule & Reminders

Daily medication timeline categorized by Morning, Afternoon, Evening, and Bedtime



Medicine Reminder Logic

- **Chronological Blocks:** Morning, Afternoon, Evening, Bedtime groupings avoid confusion.
- **Actionable Buttons:** Simple one-click "Mark as Taken" or "Snooze".
- **Audit Trail:** Displays exact timestamp and caregiver name who verified dosage.

Safety Impact

- Prevents double-dosing and forgotten medicines.
- Alerts family instantly if dose is overdue by > 30 mins.

06. Screen 5: Add / Edit Medication Form

Modal / form for family members or doctors to input medication dosage, frequency, and instructions

ElderlyCare OS > Add New Prescription

[X] Close Window

Add Medication Schedule

* Required Fields

Medicine Name:*

[Atorvastatin Lipitor]

Dosage & Unit:*

[20 mg (1 Tablet) v]

Frequency:*

[Once Daily v]

Timing Routine:*

[After Lunch v]

Time of Day:*

[01:30 PM]

Prescribed By Doctor:

[Dr. K. M. Hasan (Cardiology) v]

Current Stock / Refill Alert:

[30 Pills (Alert at 5)]

Special Instructions & Caution:

[Do not take with grapefruit juice. Take with full glass of water.]

[X] Enable SMS & App push reminder alerts

Cancel

Save & Activate Schedule

Form Field Validation

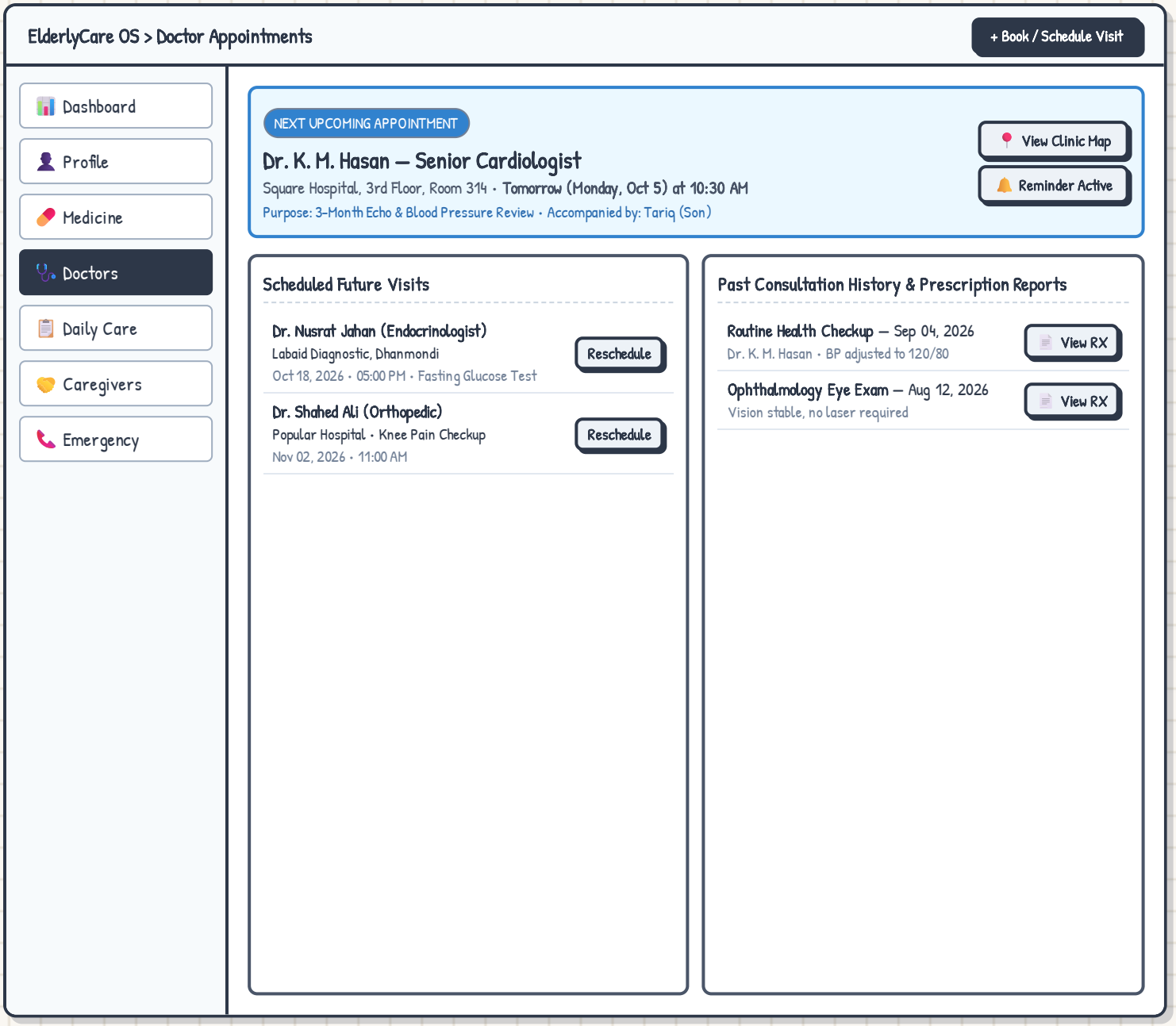
- **Dosage Safeguards:** Dropdowns ensure standard medical units (mg, ml, drops, puffs).
- **Instruction Box:** Prevents dietary contraindications (e.g. food requirements).
- **Refill Automation:** Tracks remaining pills and triggers re-order alert.

User Inputs

- Drug name and formulation
- Exact intake time & meal relation
- Notification preferences

07. Screen 6: Doctor Appointment Tracking

Healthcare appointments calendar, consultation logs, and medical booking manager



Appointment UX Highlights

- **Hero Alert Card:** Immediate visibility of nearest upcoming doctor appointment.
- **Accompaniment Info:** Identifies which family member will escort the senior.
- **Prescription Archive:** Quick link to download previous doctor notes & prescriptions.

Reduces Missed Visits

- Automated 24h and 2h SMS reminders to both caregiver and family member.

08. Screen 7: Daily Care Schedule & Activities

Routine checklist for meals, hygiene, physical mobility, hydration, and vitals recording

ElderlyCare OS > Daily Care Routine

Today's Progress: 4 of 7 Completed (57%)

- Dashboard
- Profile
- Medicine
- Doctors
- Daily Care
- Caregivers
- Emergency

- [x] **07:30 AM · Wakeup & Morning Hydration** — 1 Glass lukewarm water + lime · Verified by Nasreen (Caregiver) — Done 07:35 AM
- [x] **08:15 AM · Light Morning Exercise & Sun Exposure** — 15 min garden walk with walking stick + ankle flexion — Done 08:30 AM
- [x] **08:45 AM · Breakfast & Vitals Measurement** — Oatmeal + egg white + Papaya · BP: 124/82 mmHg — Done 09:00 AM
- [ ] **01:15 PM · Healthy Lunch Routine** — Steamed rice, boiled fish curry, mixed green vegetables (low salt) — Mark as Done
- [ ] **04:30 PM · Afternoon Hydration & Physiotherapy** — Coconut water / herbal tea + 15 min knee physiotherapy exercises — Scheduled
- [ ] **08:30 PM · Light Dinner & Warm Sponge Bath** — Vegetable soup, 2 wheat rotis, ensure room is comfortable — Scheduled

Holistic Daily Care

- **Beyond Medicine:** Incorporates hydration, diet, exercise, and mental stimulation.
- **Caregiver Handoff:** Checked items generate a daily summary report for family.
- **Large Checkboxes:** Easily tapped on mobile devices by attending staff.

Task Types

- Meal & nutrition routines
- Physical therapy / mobility
- Hygiene & vitals logging

09. Screen 8: Caregiver Information & Shift Handoff

Caregiver roster, assigned shifts, duty checklists, direct calling, and daily observation logs

ElderlyCare OS > Caregiver Directory & Shifts

+ Assign New Caregiver

- Dashboard
- Profile
- Medicine
- Doctors
- Daily Care
- Caregivers
- Emergency

Photo

CURRENTLY ACTIVE ON DUTY

Nasreen Akhtar (Certified Senior Nurse)

Day Shift: 08:00 AM – 04:00 PM • Experience: 6 Years in Geriatric Care

Agency: CareCompanion BD • Verified Police Clearance: Yes

Call: 01812-XXXXXX

Message

Caregiver Roster & Shift Schedule

Faruk Hossain (Night Shift Caregiver)
Shift: 08:00 PM – 08:00 AM • Tel: 01734-XXXXXX — Starts 8:00 PM

Morium Begum (Weekend Relief)
Fri & Sat Coverage • Tel: 01923-XXXXXX — On Leave

Shift Handoff Log & Observation Notes

Logged by Nasreen (Oct 4, 11:30 AM):
"Patient reported mild left knee soreness after walking. Applied warm compress. Appetite normal. Blood sugar at 110 mg/dL."

[Add daily handover observation note...]

Submit

Caregiver Management UX

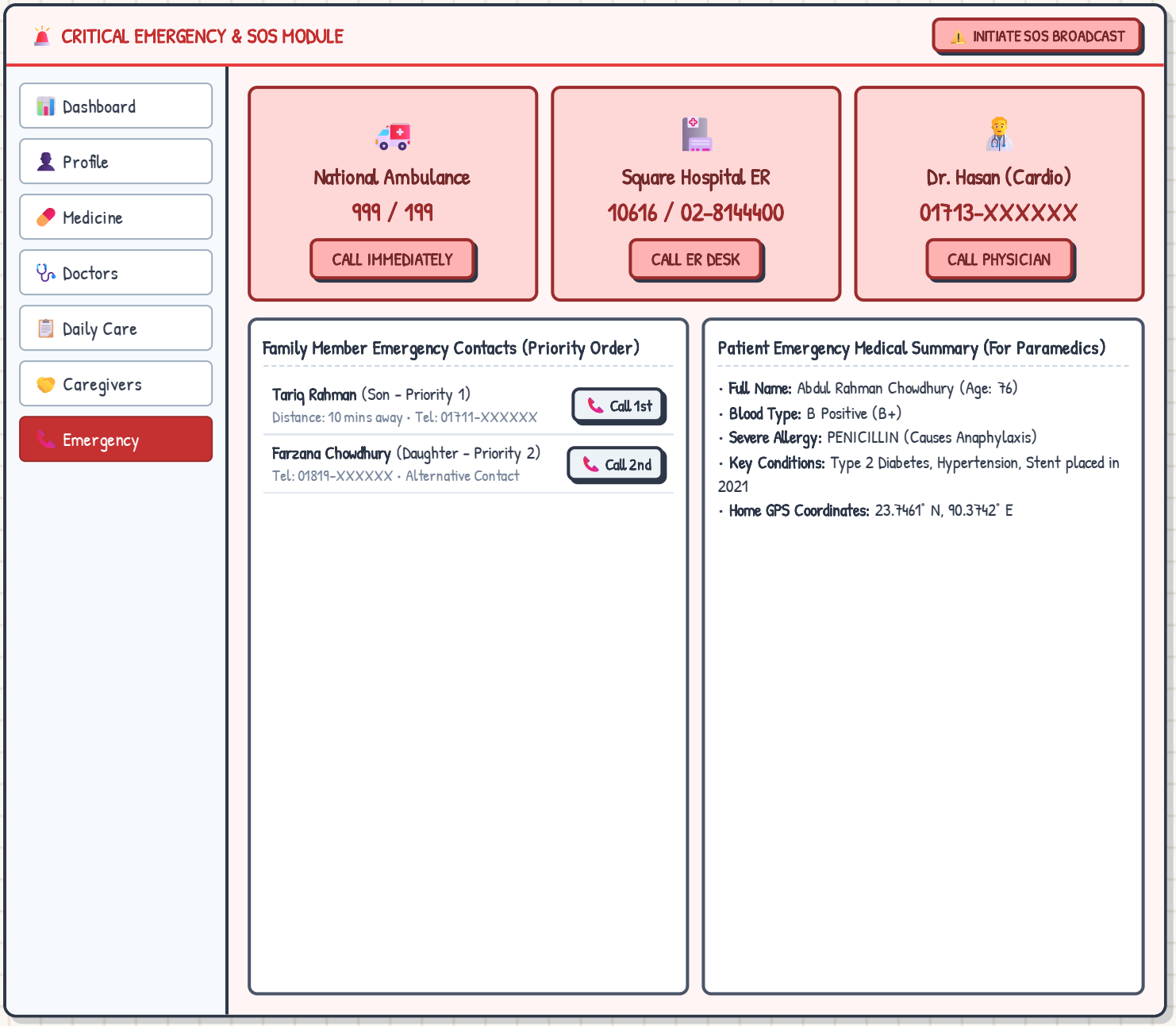
- **Direct Connectivity:** Family can phone or text the caregiver on duty with 1 tap.
- **Digital Shift Handover:** Prevents critical care omissions when shifts change.
- **Vetted Records:** Shows certification and agency contact info for transparency.

Accountability

- Shift clock-in and check-out logs.
- Mandatory shift-end handover remarks.

10. Screen 9: Emergency Contacts & Quick SOS Protocol

High-priority rapid response portal with single-tap emergency calling and hospital routing



Emergency UI Principles

- **High Affordance:** Massive, non-confusing call buttons.
- **Instant Paramedic Brief:** Blood type and Penicillin allergy instantly available to arriving medics.
- **One-Tap SMS Broadcast:** Notifies all family members simultaneously with GPS pin.

Compliance

- Meets ISO medical emergency alert standards for senior citizens.